

# Chien-Hsien Chu

FULL-STACK • DEVOPS • PLATFORM ENGINEER • 5+ YEARS WORK EXPERIENCE

☎ (+886) 975-931-286 | ✉ [chester850117@gmail.com](mailto:chester850117@gmail.com) | 📍 ChesterChuTw

## Education

### National Yang Ming Chiao Tung University (NYCU)

Hsinchu, Taiwan

M.S. IN COMPUTER SCIENCE

Sep. 2023 – Sep. 2026

- Overall GPA: 4.2/4.3 · Advisor: Prof. Chien-Chao Tseng
- Thesis: *Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions.*

## Work Experience

### ISCOM Online International Information Inc.

Taichung, Taiwan

R&D ENGINEER, R&D DEPARTMENT

Jul. 2022 – Aug. 2023

- Established a Unified Frontend Framework with **Next.js**, adopted across multiple internal products.
- Designed and implemented internal **GitLab CI/CD Pipelines** to automate build, test, and deployment workflows.
- Built and operated production **Kubernetes** clusters across on-premises and cloud environments (**AKS**).

FULL-STACK ENGINEER / TECHNICAL PROJECT MANAGER, BUSINESS UNIT

Jul. 2018 – Jun. 2022

- Architected and delivered over **45 web and enterprise projects**: E-Commerce, POS, and GIS-Based Systems.
- Built full-stack solutions with **ASP.NET Core**, **Laravel**, **Next.js**, and **Vue.js**, backed by **MSSQL** and **MySQL** databases.
- Integrated **Payment Gateways**, **LINE Official Account**, and **POS Hardware APIs** into client systems.
- Modernized legacy **.NET Framework** codebase to cross-platform, containerizable **.NET Core** architecture.
- Supported clients on **Azure Cloud** deployment, including Web Apps, VMs, and managed databases.

### Internship

Taichung, Taiwan

SOFTWARE ENGINEER INTERN

Jul. 2015 – Jun. 2018

- Three successive internships at IT companies: **ISCOM Inc.**, **BitCloud Co.**, and **RCE Power Inc.**
- Web application development, front-end performance work, and internal data-collection automation.

## Academic & Technical Experience

### 🔗 Infrastructure Automation & Management for Laboratory Environment

🔗 [source code](#)

SYSTEM ADMINISTRATOR, WIRELESS & INTERNET LAB (WINLAB), NYCU

Jul. 2024 – Sep. 2026

- Independently migrated a Kubernetes platform from **Kubespray + Terraform** to an immutable, GitOps Architecture using **Talos Linux** and **Flux CD**, including cluster bootstrap, node lifecycle automation, and PV-level data migration.
- Executed multi-application, multi-version upgrades involving rollbackable database upgrades and data migrations.
- Integrated **Velero**, **MinIO**, **Grafana Stack**, and **Keycloak** for backup, object storage, observability, and access control.
- Managed **Harvester** clusters and migrated to **Proxmox VE** for improved stability and flexibility.

### System Architecture Planning and Azure Infrastructure Consultation

TECHNICAL CONSULTANT, KOI STUDIO LTD.

Dec. 2023 – Nov. 2024

- Advised on system architecture and **Azure** infrastructure design for a software product company.
- Resolved deployment and integration issues between cloud services and external vendor APIs.

### Kubernetes Infrastructure and DevOps Workflow Development

TECHNICAL ASSISTANT, INFORMATION TECHNOLOGY SERVICE CENTER, NYCU

Sep. 2023 – Aug. 2024

- Built **Kubernetes** infrastructure for institutional systems.
- Enhanced CI/CD automation using **GitLab CI** to improve delivery reliability.

# Kubernetes Networking and Security Enhancement for Department Infrastructure

SYSTEM TEACHING ASSISTANT, CSIT, NYCU

Jul. 2024 – Sep. 2026

- Migrated networking from **Flannel** to **Calico** for fine-grained **NetworkPolicy** enforcement.
- Implemented **Istio Egress Gateway** for unified outbound traffic management.
- Evaluated **External Secrets Operator (ESO)** and **OpenBao (Vault fork)** for Kubernetes secret management.

## Multi-Cluster Management System

GRADUATE STUDENT ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Sep. 2023 – Jan. 2024

- **Multi-Cluster Management System** integrating **Karmada**, **Cilium**, and **Thanos** for orchestration and observability.

## Observability-Driven Root-Cause Analysis and Recovery Framework for Cloud-Native 5G Core Network Functions

M.S. Thesis

GRADUATE RESEARCH ASSISTANT, WIRELESS & INTERNET LAB (WINLAB), NYCU

Feb. 2026 – Sep. 2026

- Built an End-to-End Observability and Auto-Remediation Platform (**Go**, **Python**) for multi-cluster **Kubernetes**: Non-Intrusive Telemetry Collection and Correlation, Fault Detection, Root-Cause Analysis, and Guarded Remediation.
- LLM-Based RCA Agent and RAG-Based Recovery Agent: bounded investigation over live telemetry, then Knowledge Base retrieval driving **Auto** / **Approval** / **Escalation** remediation over **MCP**.
- Transparent Protocol Proxy: record-and-replay recovery of a stateful service’s in-memory state after a restart.

## Technical Skills

|                                            |                                                                                                              |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| <b>Cloud &amp; Container Orchestration</b> | Kubernetes (AKS, Talos Linux, Kubespray), Docker, Helm, Flux CD, Harbor, Karmada                             |
| <b>Observability</b>                       | Prometheus, Grafana, Loki, Tempo, Thanos, OpenTelemetry (OTLP), Alertmanager                                 |
| <b>AI &amp; LLM Systems</b>                | LLM Agents, RAG, MCP, Prompt Engineering                                                                     |
| <b>Networking &amp; Systems</b>            | Linux System Administration, IPv4 Networking, eBPF, MetallB, Traefik, Istio, Calico, Cilium                  |
| <b>Infrastructure &amp; Automation</b>     | Proxmox VE, Harvester, Terraform (OpenTofu), Ansible, Velero, GitLab CI/CD, GitOps                           |
| <b>Security &amp; Identity</b>             | Keycloak, HashiCorp Vault (OpenBao), NetworkPolicy, Cert-Manager, OAuth2 / OIDC                              |
| <b>Programming &amp; Scripting</b>         | Bash, Python, Go, C++, C# (.NET MVC/Core), PHP (Laravel), JavaScript (Vue.js / Next.js), SQL (MSSQL / MySQL) |

## Awards & Honors

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|------------------------------------------------------------------------------------|-------------|
| <b>Academic Excellence Award</b> , National Yang Ming Chiao Tung University (NYCU) | Spring 2024 |
| <b>Academic Excellence Award</b> , National Yang Ming Chiao Tung University (NYCU) | Fall 2023   |